FINAL PROJECT REPORT

SEMESTER 3, ACADEMIC YEAR: 2024-2025

CT313H: WEB TECHNOLOGIES AND SERVICES

- **Project/Application name:**
- **GitHub link (for both frontend and backend):**

https://github.com/24-25Sem3-Courses/ct313hm01-project-Hieu5980.git

- **Youtube link (demo):**

https://youtu.be/8M8IWh67J1w

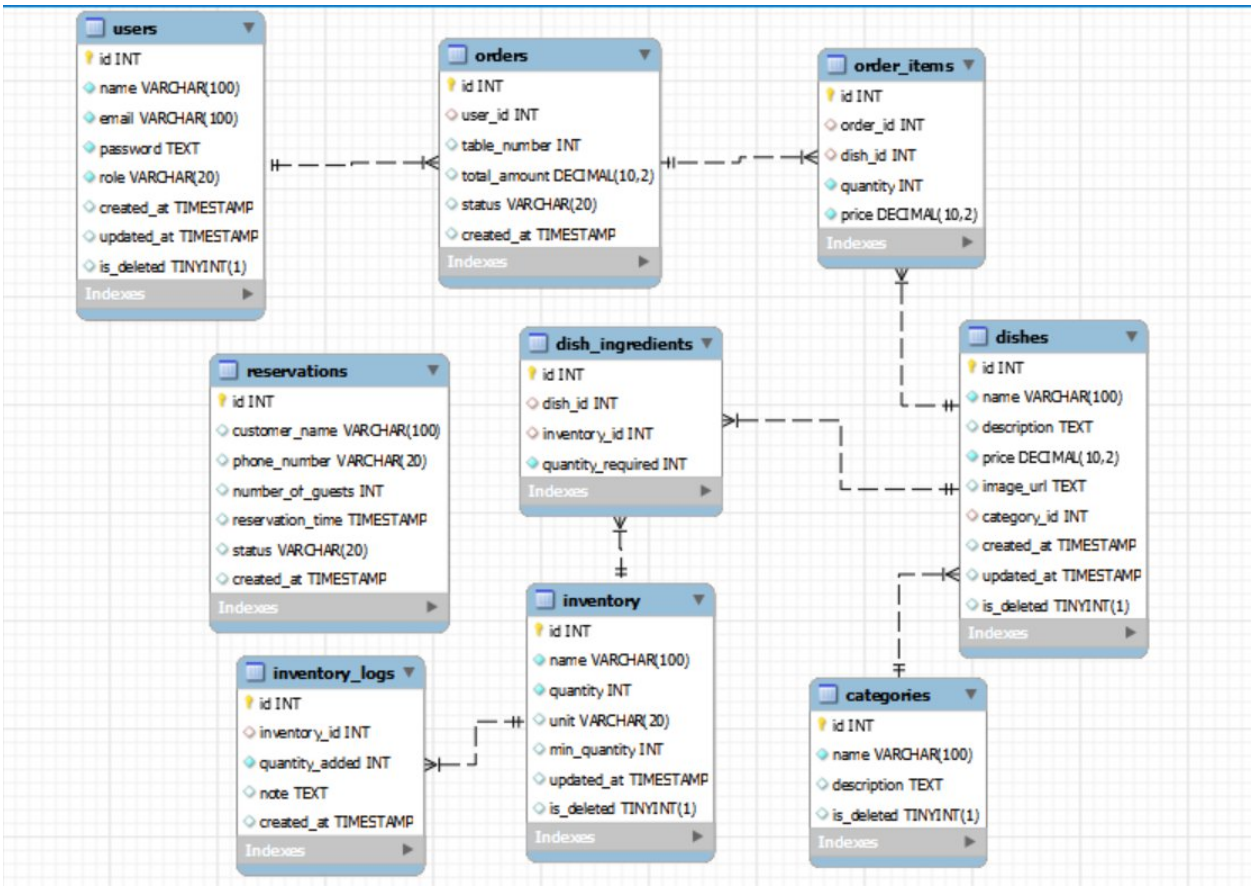
- **Student ID 1: B2205980**
- **Student Name 1: Trần Trung Hiếu**
- **Student ID 2: B2205981**
- **Student Name 2: Hồ Ngọc Hiển**
- **Class/Group Number (e.g, CT313HM01): CT313HM01**

I. Introduction

- Project/application description:

This project is a web-based restaurant management system that supports core operations of a restaurant, including managing dishes, orders, inventory, user authentication, and reservations. The backend is developed using Node.js and PostgreSQL, while the frontend is built with Vue.js. The system uses JWT for secure authentication and supports role-based access control.

- A list of tables and their structures in the database (could show a CDM/PDM diagram here):



- A task assignment sheet for each member if working in groups:

Task	Name	Student ID
Backend API	Trần Trung Hiếu	B2205980
Frontend UI	Hồ Ngọc Hiển	B2205981

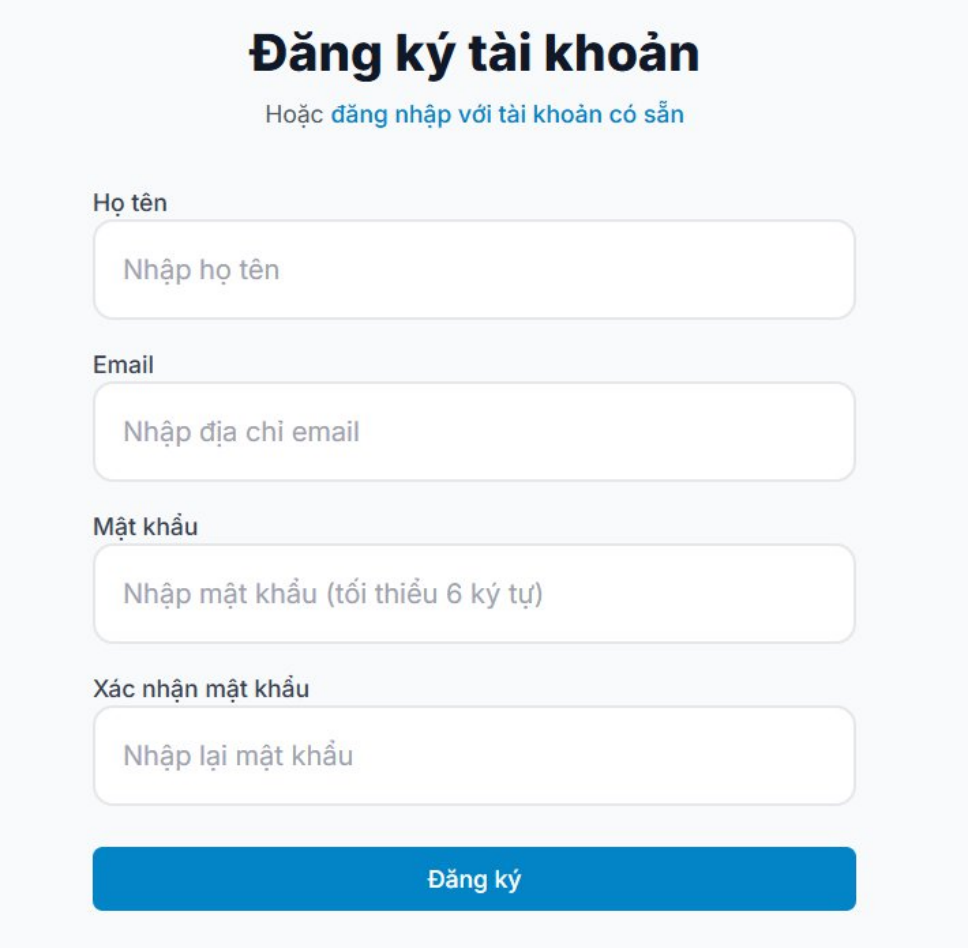
II. Details of implemented features

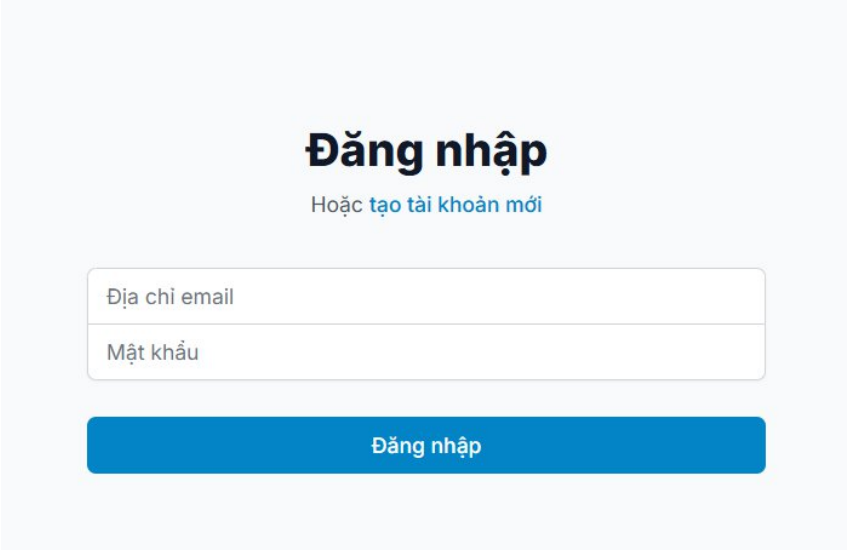
1. Feature / Application page 1: Authentication System

- **Description:**

  The authentication system handles user registration and login functionality for the restaurant management platform. This feature provides secure access control, allowing users to create accounts and authenticate themselves to access protected resources within the system.

- **Screenshots:** some screenshots of this feature / application page.





- **Implementation details:**
  - Libraries used:
    - Backend: JWT (JSON Web Tokens) for token-based authentication, bcrypt for password hashing

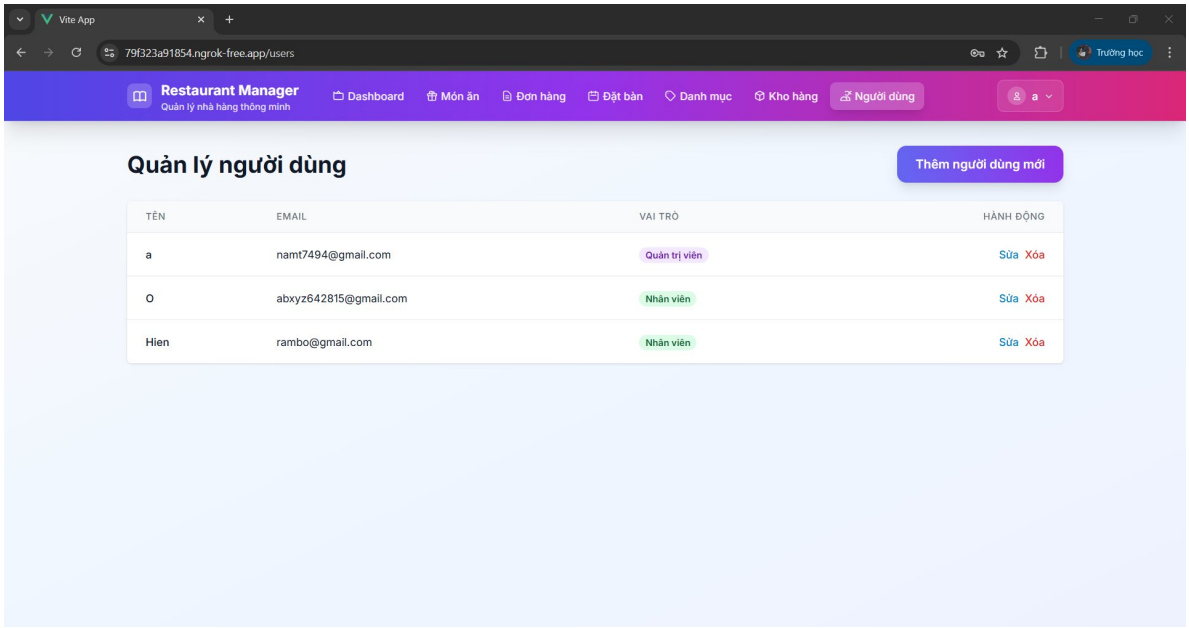
    - Frontend: Axios for HTTP requests, local storage for token management
- API Endpoints:
    - POST /api/auth/register - User registration endpoint
        - Request format: RegisterUser data (username, email, password, role)
        - Response: User object with authentication token
    - POST /api/auth/login - User authentication endpoint
        - Request format: LoginUser data (email, password)
        - Response: Authentication token and user profile data
- Table Used: users - stores user credentials, profile information, and role assignments
- Client State:
    - authToken - JWT token for authenticated requests
    - currentUser - Logged-in user profile information
    - isAuthenticated - Boolean state for authentication status
    - loginError - Error messages for failed authentication attempts

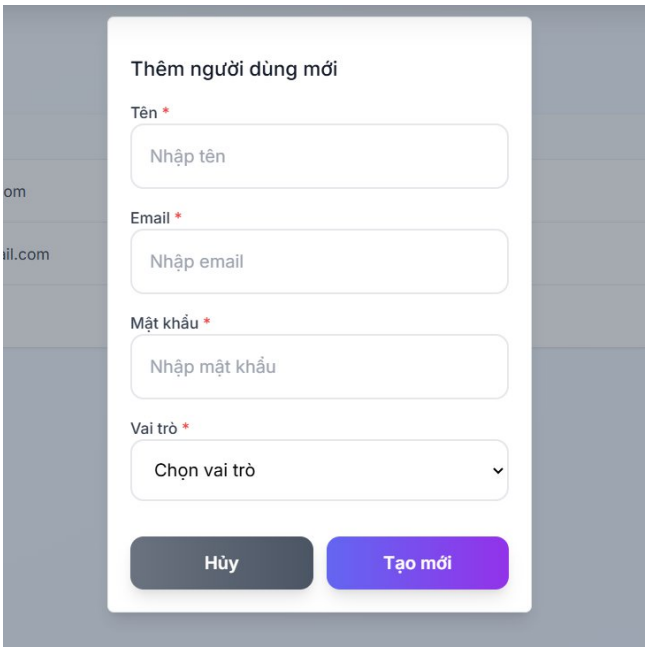
2. Feature / Application page 2: User Management System

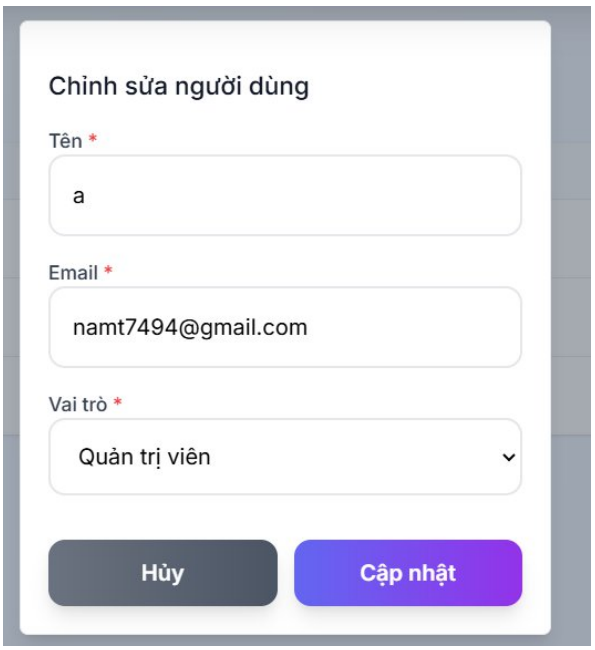
- Description:

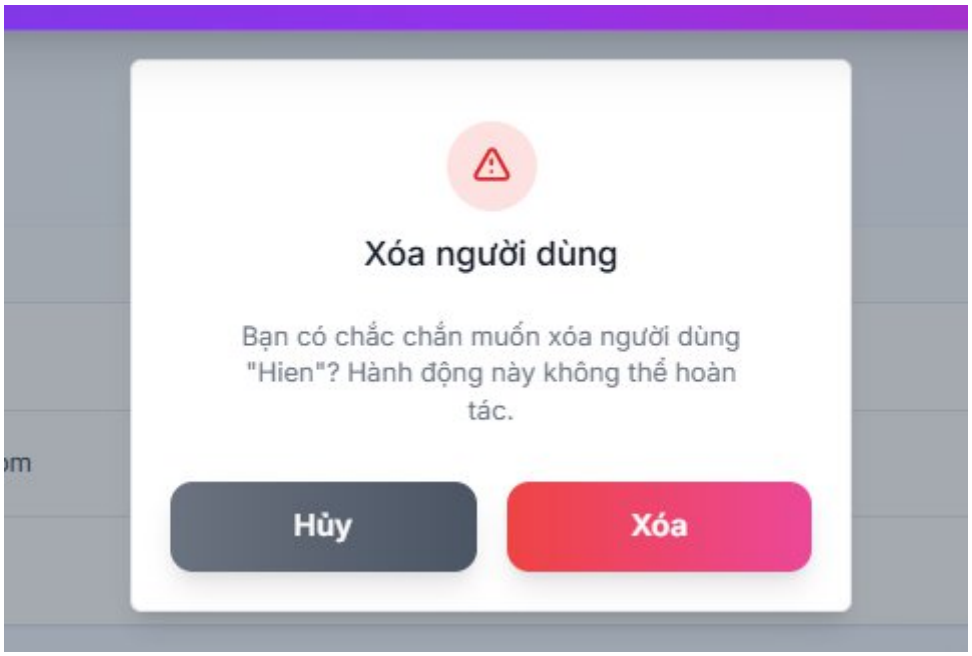
The user management feature provides comprehensive CRUD operations for managing restaurant staff and customer accounts. This system allows administrators to create, view, update, and remove user accounts, managing roles and permissions across the restaurant platform.

- **Screenshots:** some screenshots of this feature / application page.





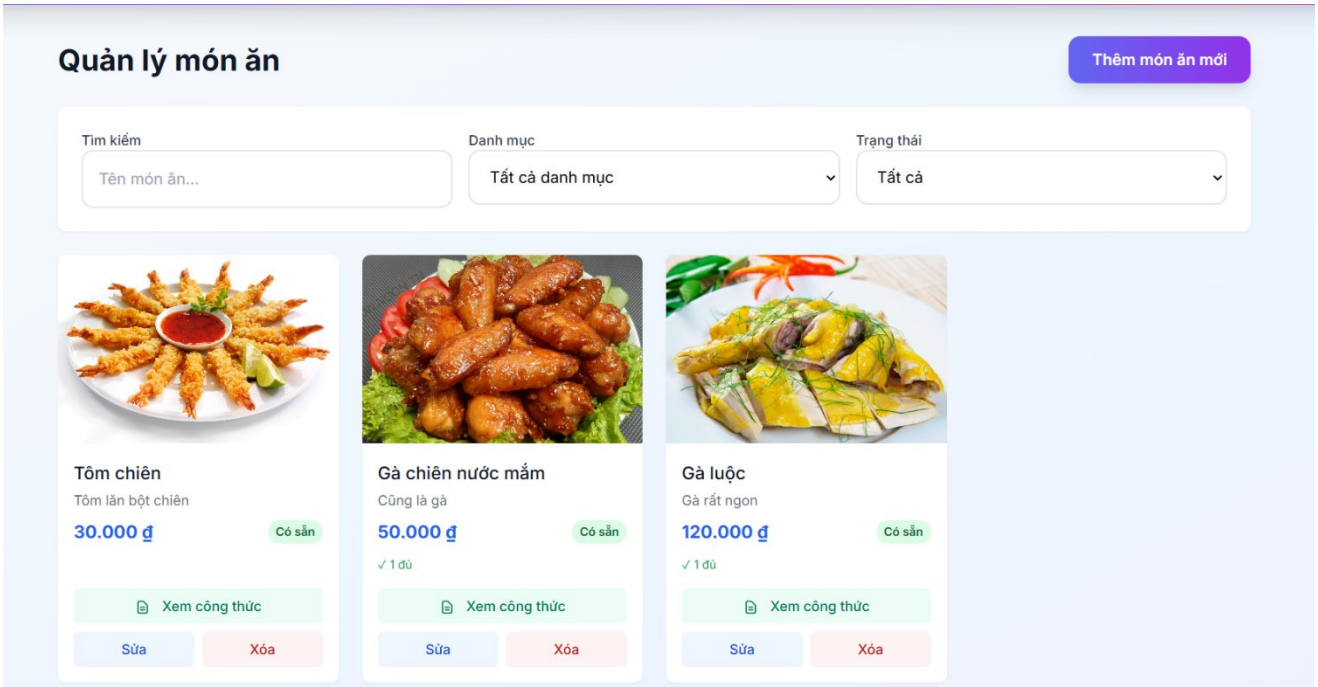




- **Implementation details:**
  - API Endpoints:
    - GET /users - Retrieve all users with pagination
      - Response: Array of User objects
    - POST /users - Create new user account
      - Request format: User schema (name, email, password, role,)
      - Response: Created user object
    - PUT /users/{id} - Update existing user information
      - Request format: Partial User schema

        - Response: Updated user object
    - DELETE /users/{id} - Remove user account
        - Response: Confirmation message
- Table Used: users - comprehensive user information
- Client State:
    - usersList - Array of all users for display
    - selectedUser - Currently selected user for editing
    - userFormData - Form state for user creation/editing
    - pagination - Current page and items per page settings

3. **Feature / Application page 3: Menu & Dish Management System**
   Description:
   The dish management system handles the restaurant's menu items, allowing admin to create, modify, and organize dishes. This feature manages dish information including names, descriptions, prices, categories, and ingredients, providing a comprehensive menu management solution.
   Screenshots: some screenshots of this feature / application page.



Thêm món ăn mới

Tên món ăn *

Nhập tên món ăn

Mô tả

Nhập mô tả món ăn

Giá *

0 VNĐ

Danh mục *

Chọn danh mục

Hình ảnh món ăn

Tải lên file hoặc kéo thả

PNG, JPG, WEBP tối đa 4MB

Hủy | Tạo món ăn

Chỉnh sửa món ăn

Tên món ăn *

Tôm chiên

Mô tả

Tôm lăn bột chiên

Giá *

30000,00 VNĐ

Danh mục *

Món khai vị

Hình ảnh món ăn

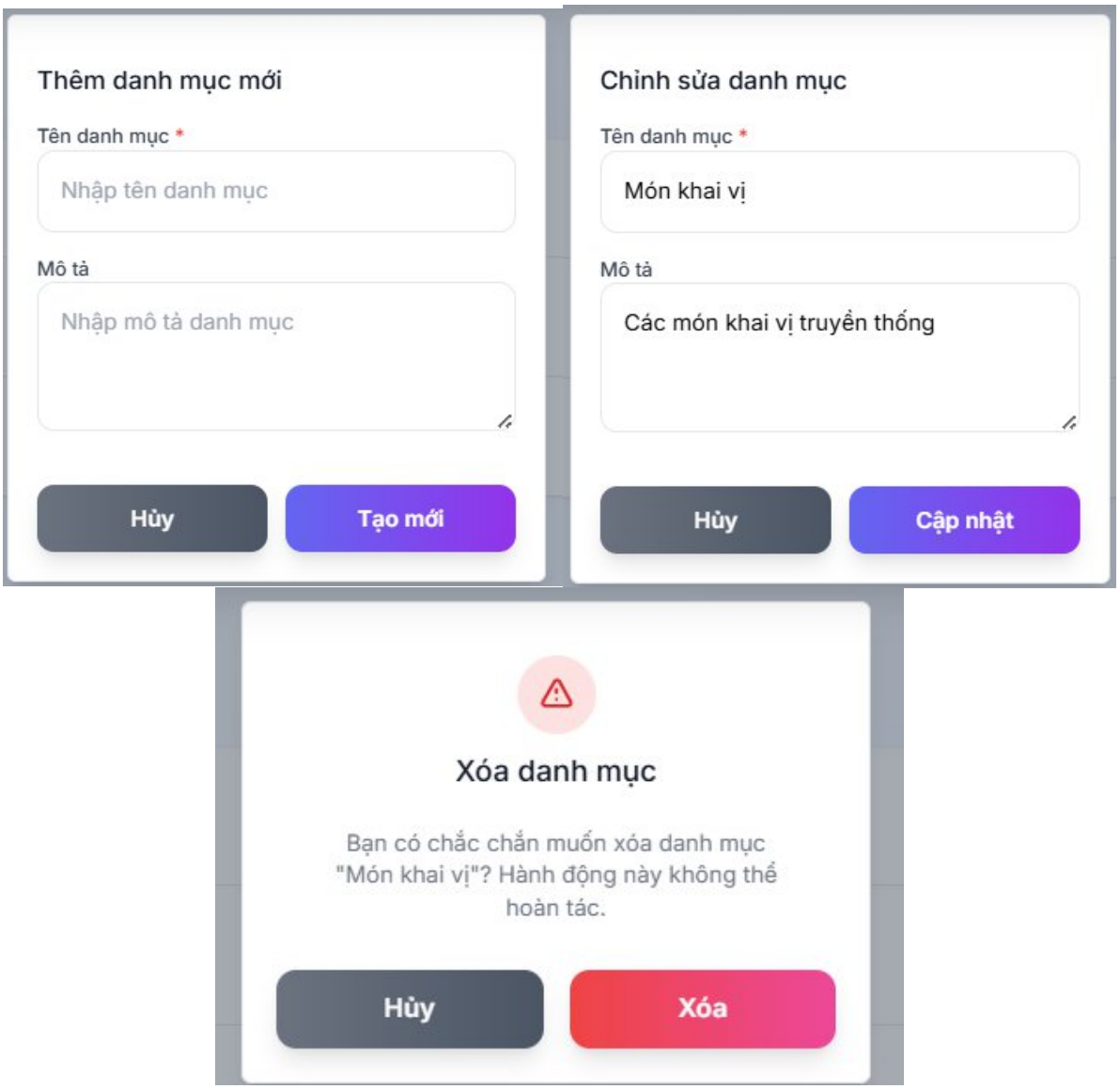
Xóa ảnh

Hủy | Cập nhật

Quản lý danh mục

Thêm danh mục mới

Món khai vị Các món khai vị truyền thống	Sửa Xóa
Món chính Các món ăn chính trong bữa ăn	Sửa Xóa
Tráng miệng Món ngọt sau bữa ăn	Sửa Xóa

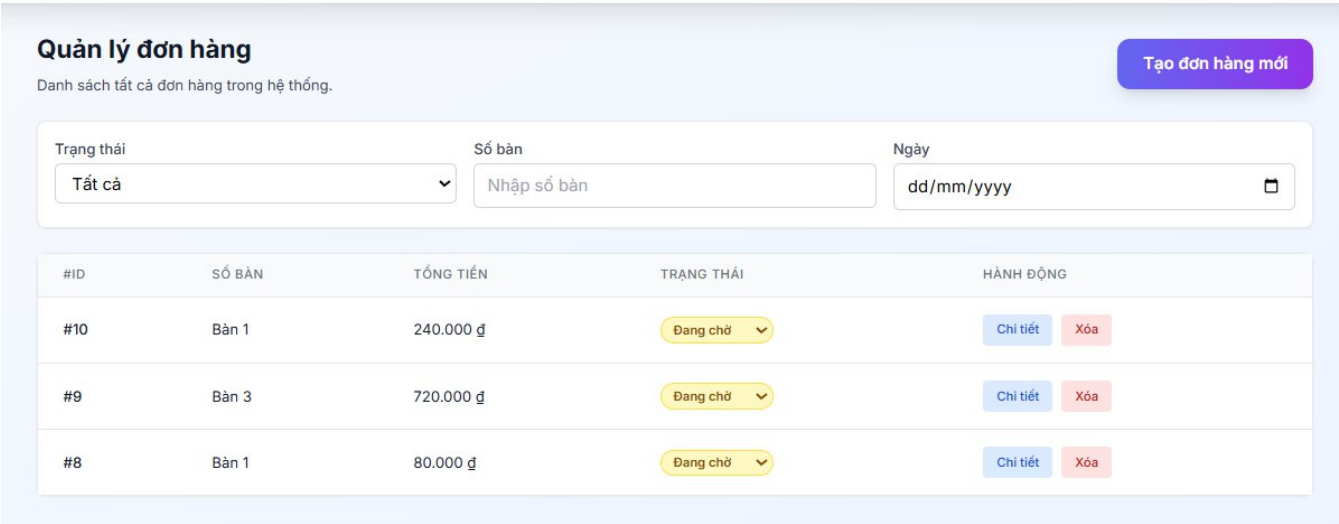


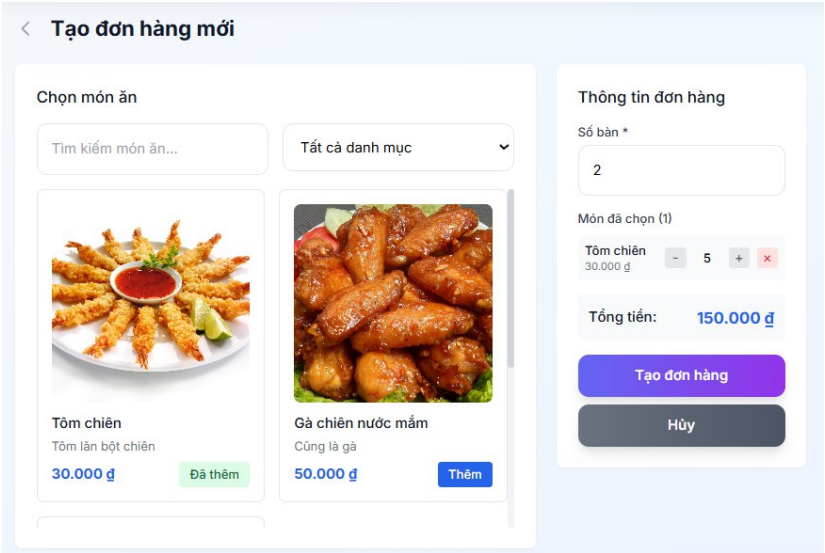
Implementation details:

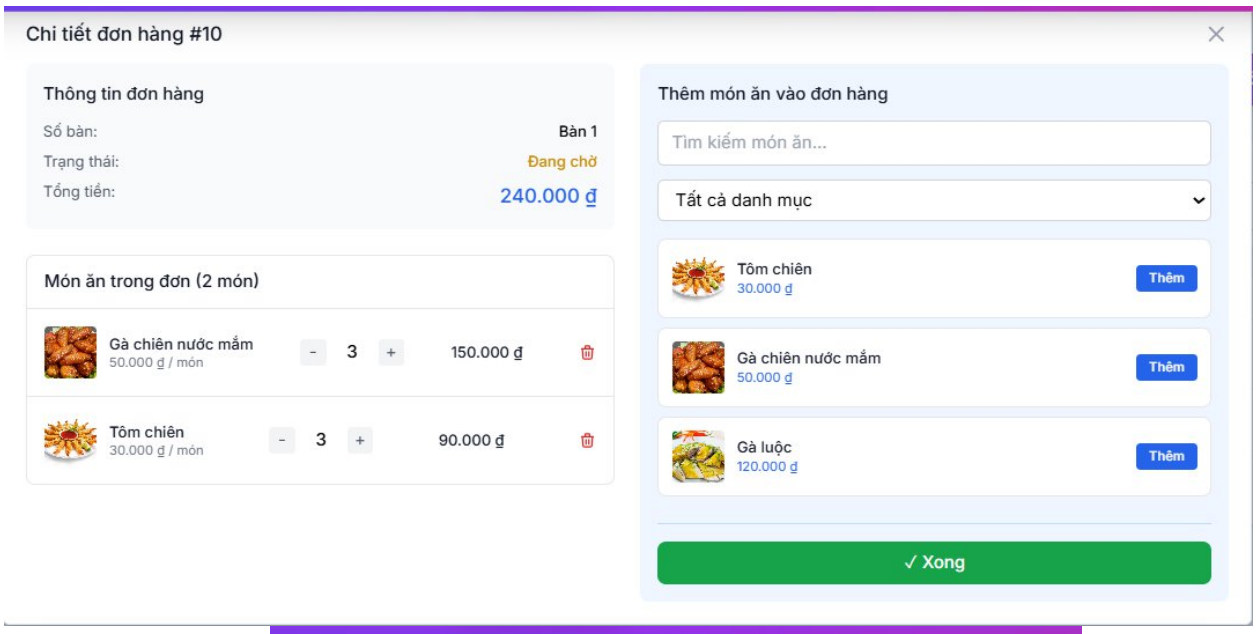
- API Endpoints:
  - GET /dishes - Retrieve all menu items with category filtering
    - Response: Array of Dish objects with category and ingredient information
  - POST /dishes - Create new dish
    - Request format: Dish schema (name, description, price, category_id, image)
    - Response: Created dish object with generated ID
  - PUT /dishes/{id} - Update dish information
    - Request format: Partial Dish schema
    - Response: Updated dish object

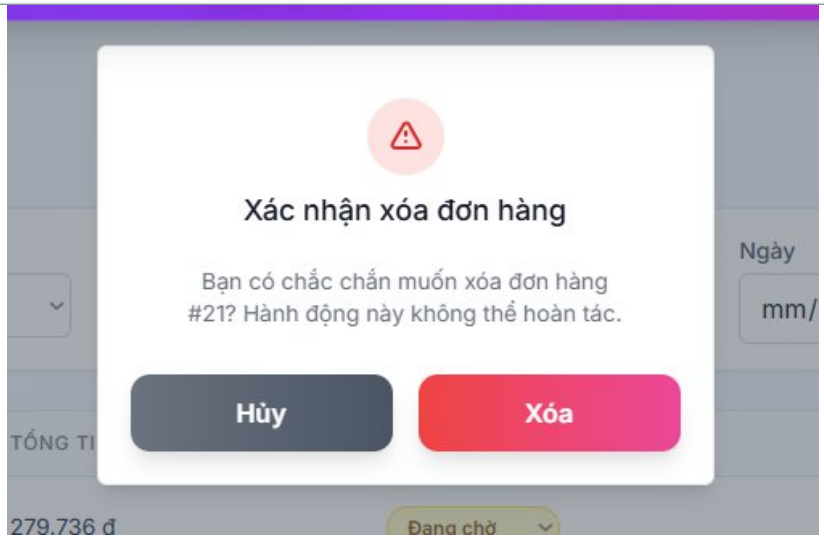
        - DELETE /dishes/{id} - Remove dish from menu
            - Response: Confirmation message
        - GET /categories - Retrieve dish categories
        - POST /categories - Create new category
        - PUT /categories/{id} - Update category information
        - DELETE /categories/{id}- Remove category
    - Table Used: dishes, categories
    - Client State:
        - DishesList - Array of all menu items
        - CategoriesList - Available dish categories
        - DishFormData - Form state for dish creation/editing
        - SelectedCategory - Current category filter
        - DishImages - Uploaded dish photos and thumbnails

4. **Feature / Application page 4: Order Management System**
   Description:
   The order management system oversees customer orders from creation to completion, including item selection, status tracking, and fulfillment. It streamlines the entire ordering process to support efficient restaurant operations.
   Screenshots: some screenshots of this feature / application page.




Tạo đơn hàng mới
Chọn món ăn
Tìm kiếm món ăn...
Tất cả danh mục
Tôm chiên
Tôm lăn bột chiên
30.000 ₫
Đã thêm
Gà chiên nước mắm
Cũng là gà
50.000 ₫
Thêm
Thông tin đơn hàng
Số bàn *
2
Món đã chọn (1)
Tôm chiên
30.000 ₫
5
Tổng tiền:
150.000 ₫
Tạo đơn hàng
Hủy


Chi tiết đơn hàng #10
Thông tin đơn hàng
Số bàn:
Bàn 1
Trạng thái:
Đang chờ
Tổng tiền:
240.000 ₫
Món ăn trong đơn (2 món)
Gà chiên nước mắm
50.000 ₫ / món
3
150.000 ₫
Tôm chiên
30.000 ₫ / món
3
90.000 ₫
Thêm món ăn vào đơn hàng
Tìm kiếm món ăn...
Tất cả danh mục
Tôm chiên
30.000 ₫
Thêm
Gà chiên nước mắm
50.000 ₫
Thêm
Gà luộc
120.000 ₫
Thêm
✓ Xong


Xác nhận xóa đơn hàng
Bạn có chắc chắn muốn xóa đơn hàng #21? Hành động này không thể hoàn tác.
Hủy
Xóa
Ngày
mm/
TỔNG TI
279.736 đ
Đang chờ

Implementation details:

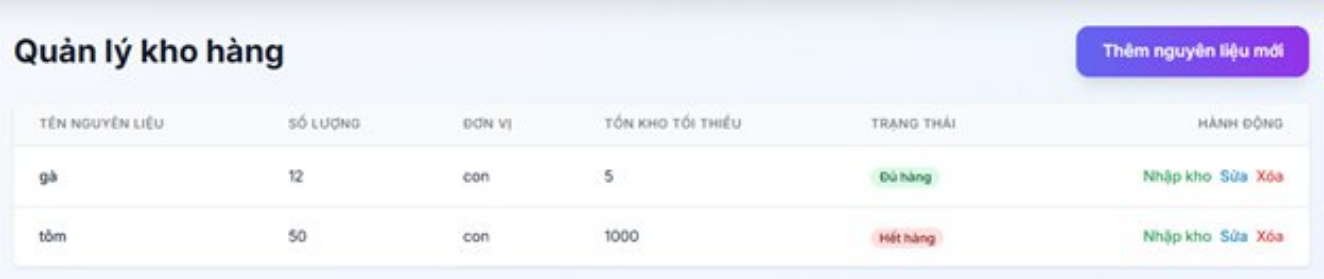
- API Endpoints:
  - GET /orders - Retrieve all orders with status filtering and pagination
    - Response: Array of Order objects with customer and item details
  - POST /orders - Create new customer order
    - Request format: Order schema (customer_id, items, table_number, total_amount)
    - Response: Created order with order number and estimated time
  - PUT /orders/{id} - Update order status or details
    - Request format: Partial Order schema (status, notes)
    - Response: Updated order object
  - DELETE /orders/{id} - Cancel order (admin only)
    - Response: Cancellation confirmation
  - GET /order-items - Retrieve order line items (admin only)
  - POST /order-items - Add items to existing order (admin only)
  - PUT /order-items/{id} - Update items from existing order (admin only)
  - DELETE /order-items/{id} - Delete items from existing order (admin only)
- Table Used: orders, order_items
- Client State:
  - OrdersList - Array of all orders with current status
  - ActiveOrders - Orders currently being processed
  - OrderDetails - Detailed view of selected order
  - OrderForm - New order creation state
  - OrderFilters - Status and date range filters

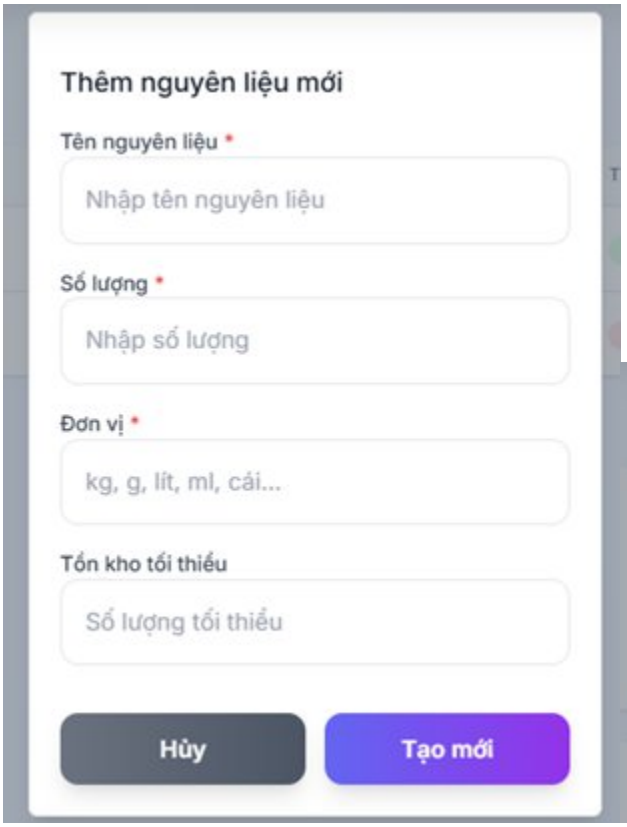
5. **Feature / Application page 5: Inventory Management System**

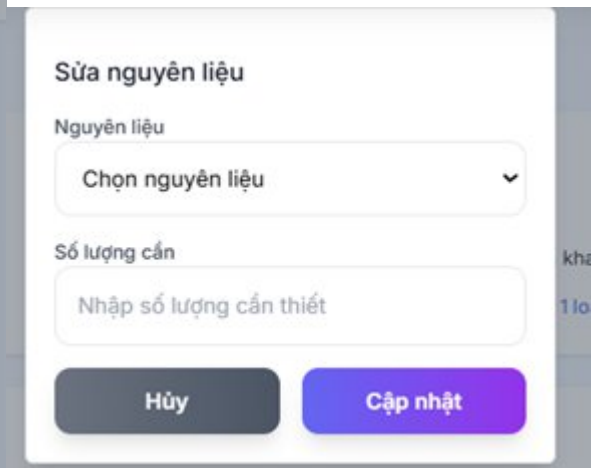
Description:

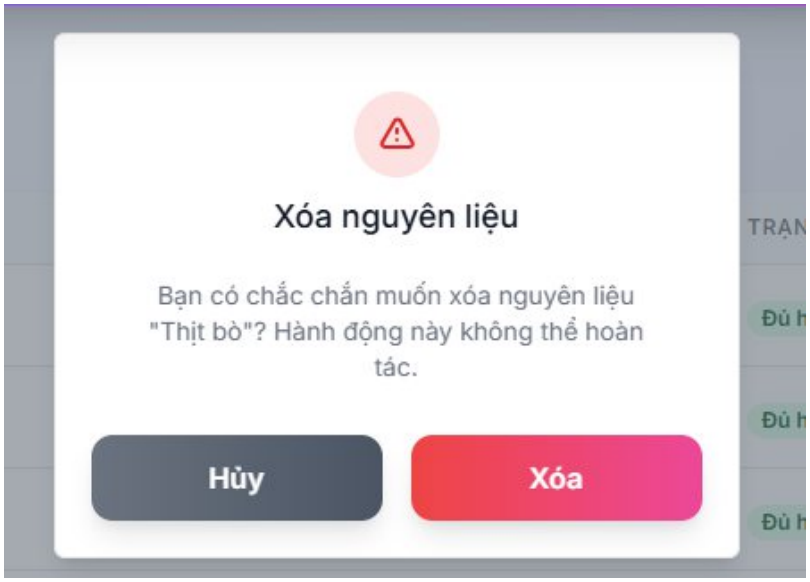
The inventory management system tracks restaurant ingredients and supplies, usage patterns, and reorder points. This feature provides comprehensive inventory control with automated alerts for low stock and detailed logging of inventory transactions

Screenshots: some screenshots of this feature / application page


Quản lý kho hàng
Thêm nguyên liệu mới
TÊN NGUYÊN LIỆU
SỐ LƯỢNG
ĐƠN VỊ
TỒN KHO TỐI THIỂU
TRẠNG THÁI
HÀNH ĐỘNG
gà
12
con
5
Đủ hàng
Nhập kho Sửa Xóa
tôm
50
con
1000
Hết hàng
Nhập kho Sửa Xóa


Thêm nguyên liệu mới
Tên nguyên liệu *
Nhập tên nguyên liệu
Số lượng *
Nhập số lượng
Đơn vị *
kg, g, lít, ml, cái...
Tồn kho tối thiểu
Số lượng tối thiểu
Hủy
Tạo mới

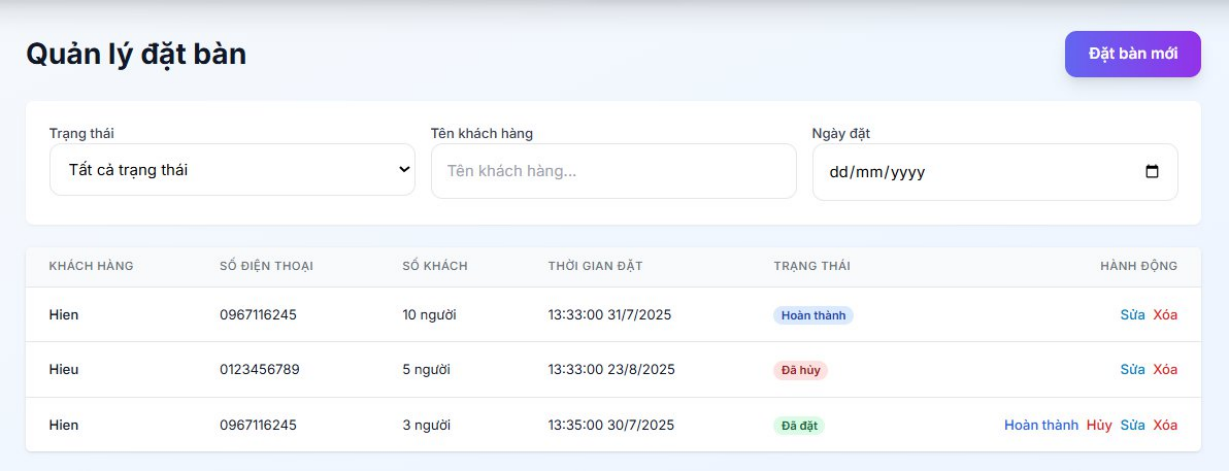

Sửa nguyên liệu
Nguyên liệu
Chọn nguyên liệu
Số lượng cần
Nhập số lượng cần thiết
Hủy
Cập nhật

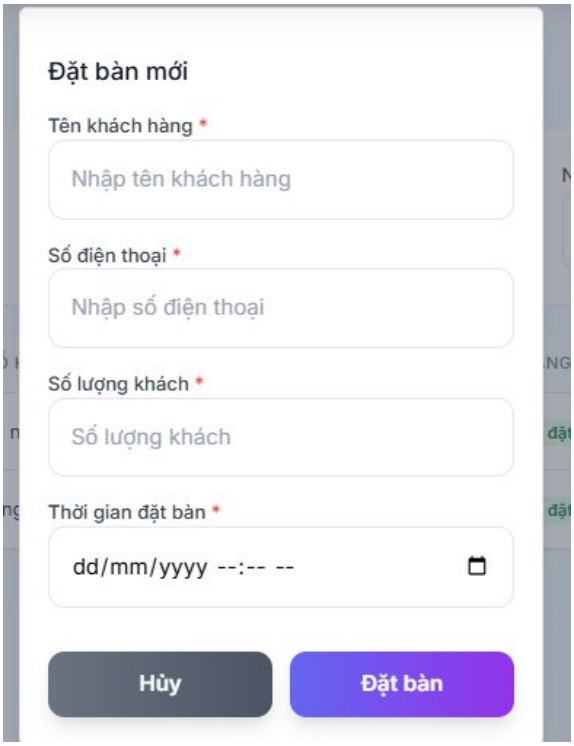

Xóa nguyên liệu
Bạn có chắc chắn muốn xóa nguyên liệu "Thịt bò"? Hành động này không thể hoàn tác.
Hủy
Xóa

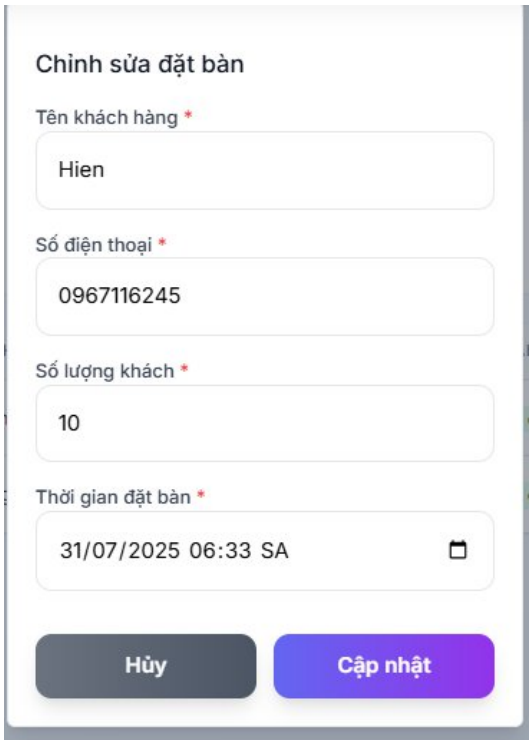
Implementation details:

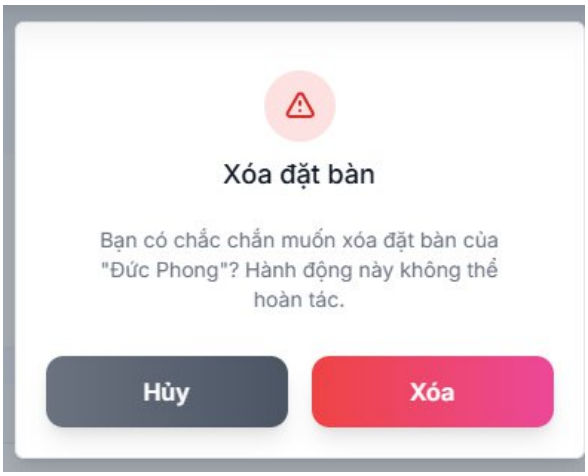
- API Endpoints:
  - GET /inventory - Retrieve all inventory items with current stock levels
    - Response: Array of Inventory objects with usage statistics
  - POST /inventory - Add new inventory item
    - Request format: Inventory schema (name, quantity, unit, supplier, reorder_point)
    - Response: Created inventory item
  - PUT /inventory/{id} - Update inventory quantities or details
    - Request format: Partial Inventory schema
    - Response: Updated inventory item
  - DELETE /inventory/{id} - Remove inventory item
  - GET /inventory-logs - Retrieve inventory transaction history
  - POST /inventory-logs - Log inventory transactions
- Table Used: inventory, inventory_logs
- Client State:
  - InventoryItems - Array of all inventory items
  - LowStockItems - Items below reorder point
  - InventoryLogs - Transaction history
  - StockAlerts - Active low stock notifications
  - InventoryForm - Form state for adding/editing items
  - UsageAnalytics - Consumption patterns and trends
  -

6. **Feature / Application page 6: Reservation Management**
   Description:
   The reservation management system handles customer table bookings, allowing customers to reserve tables and staff to manage dining room capacity. This feature includes reservation scheduling, table assignment, and reservation status tracking.
   Screenshots: some screenshots of this feature / application page.


Quản lý đặt bàn
Đặt bàn mới
Trạng thái
Tất cả trạng thái
Tên khách hàng
Tên khách hàng...
Ngày đặt
dd/mm/yyyy
KHÁCH HÀNG
SỐ ĐIỆN THOẠI
SỐ KHÁCH
THỜI GIAN ĐẶT
TRẠNG THÁI
HÀNH ĐỘNG
Hien
0967116245
10 người
13:33:00 31/7/2025
Hoàn thành
Sửa Xóa
Hieu
0123456789
5 người
13:33:00 23/8/2025
Đã hủy
Sửa Xóa
Hien
0967116245
3 người
13:35:00 30/7/2025
Đã đặt
Hoàn thành Hủy Sửa Xóa


Đặt bàn mới
Tên khách hàng *
Nhập tên khách hàng
Số điện thoại *
Nhập số điện thoại
Số lượng khách *
Số lượng khách
Thời gian đặt bàn *
dd/mm/yyyy --:-- --
Hủy
Đặt bàn


Chỉnh sửa đặt bàn
Tên khách hàng *
Hien
Số điện thoại *
0967116245
Số lượng khách *
10
Thời gian đặt bàn *
31/07/2025 06:33 SA
Hủy
Cập nhật


Xóa đặt bàn
Bạn có chắc chắn muốn xóa đặt bàn của "Đức Phong"? Hành động này không thể hoàn tác.
Hủy
Xóa

Implementation details:

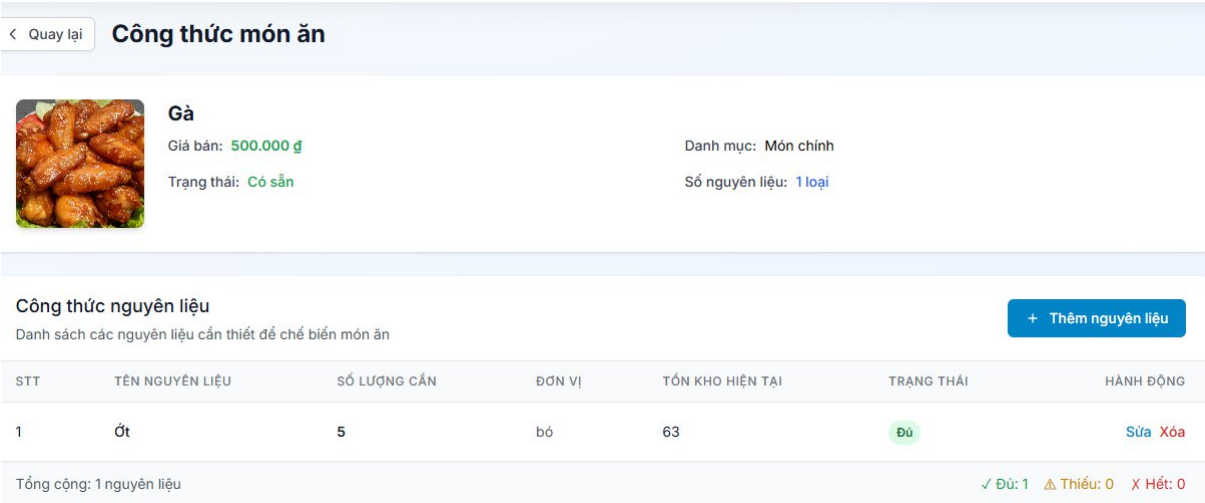
- API Endpoints:
  - GET /reservations - Retrieve all reservations with date filtering
    - Response: Array of Reservation objects with customer and table details
  - POST /reservations - Create new table reservation
    - Request format: Reservation schema (customer_info, date, time, party_size, special_requests)
    - Response: Confirmed reservation with table assignment
  - PUT /reservations/{id} - Update reservation details or status
    - Request format: Partial Reservation schema
    - Response: Updated reservation object
  - DELETE /reservations/{id} - Cancel reservation
    - Response: Cancellation confirmation
- Table Used: reservations
- Client State:
  - ReservationsList - Array of all reservations
  - ReservationForm - Booking form state
  - CalendarView - Date selection and schedule display

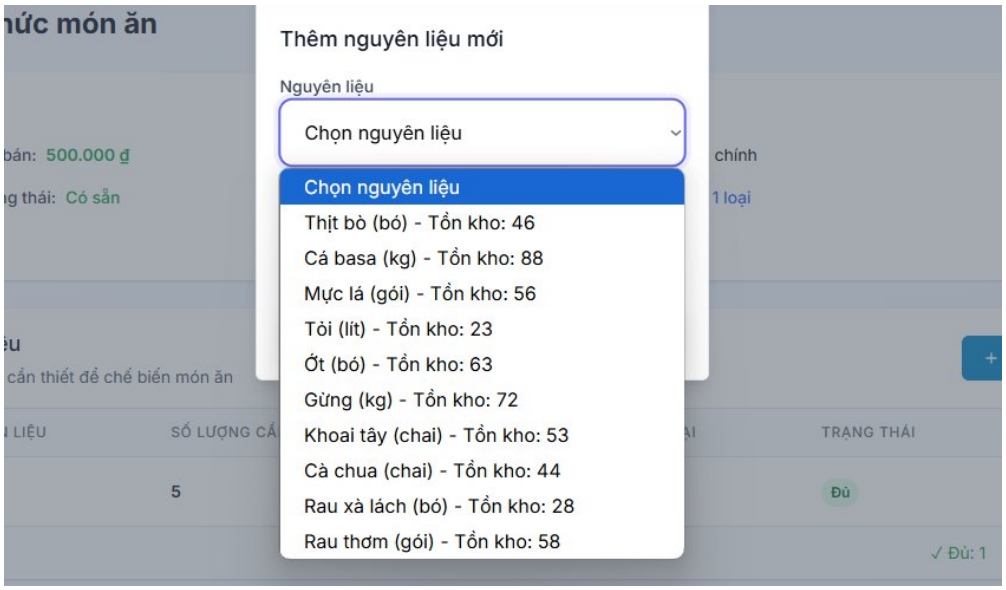
7. **Feature / Application page 7: Recipe & Ingredient Tracking**

Description:

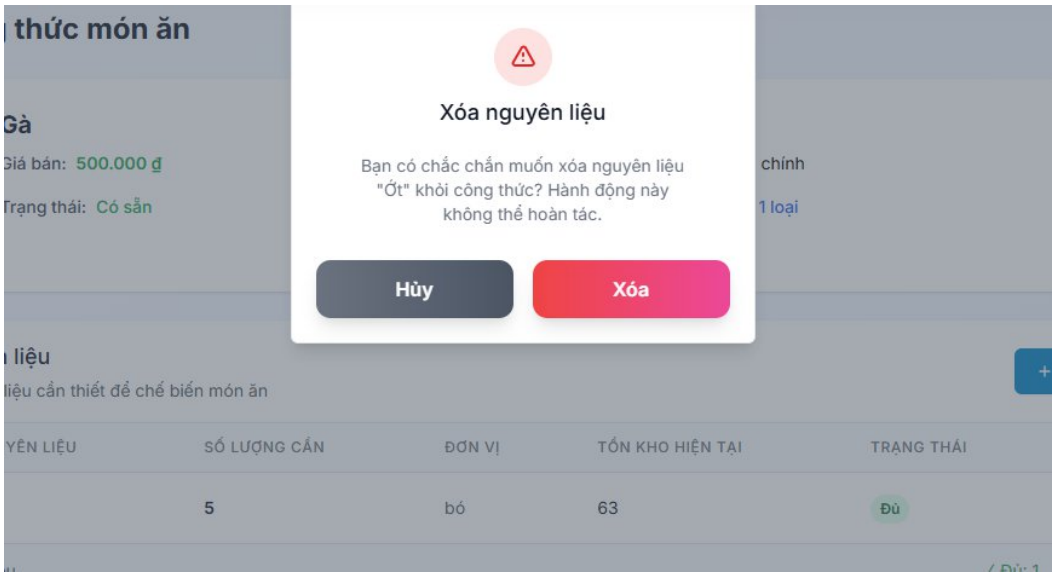
The dish ingredients system manages the relationship between menu items and inventory ingredients, tracking recipe compositions and calculating ingredient costs. This feature enables precise inventory management and cost analysis for each dish.

Screenshots: some screenshots of this feature / application page.




Thêm nguyên liệu mới
Nguyên liệu
Chọn nguyên liệu
Chọn nguyên liệu
Thịt bò (bó) - Tồn kho: 46
Cá basa (kg) - Tồn kho: 88
Mực lá (gói) - Tồn kho: 56
Tỏi (lít) - Tồn kho: 23
Ớt (bó) - Tồn kho: 63
Gừng (kg) - Tồn kho: 72
Khoai tây (chai) - Tồn kho: 53
Cà chua (chai) - Tồn kho: 44
Rau xà lách (bó) - Tồn kho: 28
Rau thơm (gói) - Tồn kho: 58
TRẠNG THÁI
Đủ
✓ Đủ: 1

Sửa nguyên liệu
Nguyên liệu
Ớt (bó) - Tồn kho: 63
Số lượng cần
5
Hủy
Cập nhật



Implementation details:

- API Endpoints:
  - GET /dish-ingredients - Retrieve ingredient relationships
    - Response: Array of DishIngredient objects with quantities and costs
  - POST /dish-ingredients - Link ingredients to dishes
    - Request format: DishIngredient schema (dish_id, ingredient_id, quantity, unit)
    - Response: Created ingredient relationship
  - PUT /dish-ingredients/{id} - Update ingredient quantities
  - DELETE /dish-ingredients/{id} - Remove ingredient from recipe
- Table Used: dish_ingredients
- Client State:
  - DishIngredients - Recipe compositions
  - IngredientCosts - Real-time cost calculations
  - RecipeBuilder - Recipe creation interface state